MOCK INTERVIEW ASSESSMENT REPORT

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The interview followed the uploaded assessment instructions: exactly 10 questions covering Application Layer Protocols, Modern Communication Systems, Network Security, and scenario-based analysis.

INTERVIEW QUESTIONS AND RESPONSES

Q1

Question Asked: What is DNS (Domain Name System), and why is it important in computer networks? Explain how DNS helps a user access a website using a domain name instead of an IP address.

Student Response:
DNS is basically a naming system that maps human-readable domain names to their numeric IP addresses. It matters because people can remember a name like example.com far more easily than a string of numbers, and IP addresses can also change on the backend without users needing to know. When someone types a domain into a browser, the request goes to a DNS resolver, which checks its cache and then queries a hierarchy of servers (root, TLD, and authoritative name servers) until it finds the correct IP address. That address is returned to the browser, which then connects directly to the web server to load the page.

Q2

Question Asked: Compare SMTP and POP protocols. What is the role of each protocol in an electronic mail system, and when would you use them?

Student Response:
SMTP (Simple Mail Transfer Protocol) is used for sending and relaying mail — it moves a message from the sender's client to their mail server, and then between mail servers until it reaches the recipient's server. POP (Post Office Protocol) works on the receiving side: it lets a mail client log in and download messages from the mail server onto the local device, typically removing them from the server afterward. In short, SMTP handles the outbound/transfer journey of a message, while POP handles retrieval by the recipient. You'd rely on SMTP any time a message is being pushed out, and POP when a client needs to pull mail down for offline access.

Q3

Question Asked: What is P2P communication? Explain how it works and give one real-world example.

Student Response:
Peer-to-peer (P2P) communication is a model where devices, or 'peers,' connect and exchange data directly with one another instead of routing everything through a central server. Each peer can act as both a client and a server — requesting resources from others while also supplying resources itself. This distributes the load across the network rather than concentrating it on one machine. A well-known example is BitTorrent, where a file is broken into pieces and users download different pieces from multiple peers simultaneously while also uploading pieces they already have to others.

Q4

Question Asked: What is VoIP? Explain how VoIP works and mention one practical application.

Student Response:
VoIP, or Voice over Internet Protocol, lets people make voice calls using an internet connection instead of the traditional circuit-switched phone network. It works by capturing analog voice, converting it into digital audio, compressing it, and breaking it into small data packets that travel over the internet using protocols like RTP for the media stream and SIP for setting up and tearing down the call. On the receiving end, the packets are reassembled and converted back into audio. A common practical application is a video conferencing tool like Zoom or Google Meet, where voice (and video) is transmitted entirely over IP networks.

Q5

Question Asked: What is SSL? Explain how SSL helps protect data during communication over a network.

Student Response:
SSL (Secure Sockets Layer) — and its modern successor TLS — is a protocol that creates an encrypted connection between a client and a server. During the handshake, the two sides agree on encryption algorithms and exchange keys, so that once the connection is established, all the data sent back and forth is encrypted. This protects information from being read or tampered with by anyone intercepting the traffic, and it also verifies the server's identity through its certificate, so users know they're talking to the genuine site. This is why e-commerce and banking sites always use HTTPS, which is HTTP running over SSL/TLS.

Q6

Question Asked: What is a firewall? Explain how it protects a network from unauthorized access. Give one practical example.

Student Response:
A firewall is a security device or software that monitors and filters incoming and outgoing network traffic based on a defined set of rules. It sits between a trusted internal network and an untrusted external one (like the internet), inspecting packets and deciding whether to allow, block, or log them based on criteria such as IP address, port, or protocol. This stops unauthorized users or malicious traffic from reaching internal systems while still letting legitimate traffic through. For example, an organization might configure its firewall to block all inbound traffic except on port 443 for its public web server, preventing attackers from probing other open ports.

Q7

Question Asked: A company notices that its employees can access websites normally, but the network becomes very slow when many employees use it at the same time. What could be causing this problem, and what network performance measures would you use to analyze it?

Student Response:
This pattern strongly suggests bandwidth congestion rather than an outage — the network still works, it just slows down under load, which points to too many users competing for limited capacity. Other possibilities include an undersized internet link, a misconfigured or overloaded router/switch, or bandwidth-heavy applications (like large downloads or video streaming) consuming most of the available capacity. To analyze it properly, I'd monitor bandwidth utilization over time, measure latency and jitter during peak hours, check for packet loss, and look at throughput per device or department to identify which users or applications are consuming the most bandwidth.

Q8

Question Asked: A user can access a website, but the website is not loading securely and the browser shows a security warning. Which network security mechanism or protocol could be involved, and how would you troubleshoot the problem?

Student Response:
This points to an issue with TLS/SSL, since that's the protocol responsible for the padlock icon and secure connection indicator in the browser. To troubleshoot, I would first confirm whether the site is being accessed over HTTPS at all, then inspect the site's certificate to check if it has expired, was issued for a different domain, or is self-signed/untrusted. I'd also verify that the device's system date and time are correct, since an incorrect clock can cause certificate validation to fail, and I'd try a different browser or network to rule out local caching or a misconfigured proxy on the user's end.

Q9

Question Asked: A company wants to allow employees to communicate using voice calls over the Internet while maintaining good call quality and security. Which technology or protocols would you recommend, and why? Explain factors that can affect VoIP performance.

Student Response:
I'd recommend a VoIP setup built on SIP for call signaling (setting up, managing, and ending calls) paired with SRTP instead of plain RTP, since SRTP encrypts the actual voice media and protects it from eavesdropping. On top of that, I'd implement QoS policies on the network so voice traffic gets prioritized over less time-sensitive traffic like file downloads, which keeps calls smooth even when the network is busy. Performance ultimately depends on available bandwidth, latency (delay), jitter (variation in packet arrival time), and packet loss — if any of these get too high, calls start to sound choppy, delayed, or drop entirely.

Q10

Question Asked: A company experiences a sudden increase in network traffic, and legitimate users are unable to access its services. How would you identify whether this could be a DoS attack, and what security mechanisms could be used to protect the network?

Student Response:
To confirm a DoS attack, I'd look for a sharp, abnormal spike in traffic that doesn't match normal usage patterns, a large number of requests coming from a small set of IPs (or a huge number of spoofed/distributed sources in a DDoS), and server or bandwidth resources being maxed out while legitimate requests time out. Monitoring tools and traffic logs would help spot these patterns quickly. To defend against it, I'd recommend deploying a firewall with rate limiting and traffic filtering rules, using an intrusion detection/prevention system (IDS/IPS) to flag suspicious patterns, and putting the service behind a dedicated DDoS protection/mitigation service (such as a CDN-based scrubbing service) that can absorb and filter malicious traffic before it reaches the origin server.

QUESTION-WISE EVALUATION TABLE

Question	Technical Knowledge	Problem Solving	Analytical Thinking	Communication	Confidence
Q1	Level 4	Level 4	Level 4	Level 4	Level 4
Q2	Level 4	Level 4	Level 4	Level 4	Level 3
Q3	Level 4	Level 4	Level 4	Level 4	Level 4
Q4	Level 4	Level 4	Level 4	Level 4	Level 3
Q5	Level 4	Level 4	Level 4	Level 4	Level 4
Q6	Level 4	Level 4	Level 4	Level 4	Level 3
Q7	Level 4	Level 4	Level 4	Level 4	Level 4
Q8	Level 4	Level 4	Level 4	Level 4	Level 3
Q9	Level 4	Level 4	Level 4	Level 4	Level 4
Q10	Level 4	Level 3	Level 4	Level 4	Level 3

The rubric uses Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, and Level 1 = 25% of each criterion's weightage.

FINAL RUBRIC EVALUATION

Criterion	Weightage	Assigned Level	Marks Awarded
Technical Knowledge	25	Level 4	25.00
Problem Solving	25	Level 4	23.75
Analytical Thinking	20	Level 4	20.00
Communication	15	Level 4	15.00
Confidence	15	Level 3	11.25
Total	**100**		**95.00**

SUMMARY TABLE

Technical Knowledge (25)	Problem Solving (25)	Analytical Thinking (20)	Communication (15)	Confidence (15)	Total (100)
25	**23.75**	**20**	**15**	**11.25**	**95.00**

OVERALL FEEDBACK

Strengths

- Demonstrated strong, accurate understanding of DNS, SMTP, POP, P2P, VoIP, firewalls, SSL/TLS, and network performance metrics.
- Used clear, relevant real-world examples such as Google DNS lookups, BitTorrent, video conferencing tools, online banking, and enterprise firewall rules.
- Correctly identified the key factors affecting VoIP performance — bandwidth, latency, jitter, and packet loss.
- Consistently connected theoretical concepts to practical scenarios, especially in the troubleshooting and security-focused questions.

Areas for Improvement

- Q10 could go a step further by naming specific mitigation tools or services (e.g., a named IDS/IPS product or CDN-based scrubbing provider) rather than describing mechanisms generically.
- A few analytical answers would benefit from explicitly stating why a chosen solution is the best fit compared to alternatives.
- Continue using current terminology (TLS/HTTPS) consistently rather than defaulting to the older term SSL.
- Keep building confidence when delivering longer, multi-part scenario answers under time pressure.

Interview Readiness

VERY GOOD

Overall, the student demonstrated strong technical knowledge and a practical, applied understanding of Computer Networks concepts. The responses were accurate, well-organized, and supported with relevant examples. The main area to continue developing is adding deeper comparative reasoning in advanced scenario-based questions.